Tree Ring Climate Detective

Read rings to reconstruct past climate

At a glance: Read the rings to reconstruct the climate events a tree lived through. Plan about 80 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: tree rings, proxy data, paleoclimate, pattern recognition (Understand).
- Read the rings in order and infer past climate events, citing the exact rings as evidence (Apply).
- Weigh the ring evidence to decide which climate story best fits the sequence (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- printed ring cards: one set per team (4, plus 1 spare), laminated for reuse
- answer boards: one per team (4, plus 1 spare), reusable
- wood cookies (optional): one sanded slice per team if used (4, plus 1 spare); the printed cards are the primary medium
- timers: one per team (4) or the projected room timer
- claim-evidence cards: a stack per team (about 5 to 7 each), printed on the shared cardstock line

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

01 Set up 4 stations, one per team, each with one ring-card set, one answer board, one timer, a stack of claim-evidence cards, markers, the ring-code legend, and (if used) one sanded wood cookie, sorted and ready.

02 Build the answer key for each ring-card set (the correct climate-event sequence) and stage it with the instructor, plus the score slips, before play begins.

03 No PPE is needed: this is a dry card-and-board game with no soil, water, or chemicals. If you use real wood cookies, sand them smooth and inspect for splinters and loose bark beforehand, and skip them for any student with a wood-dust sensitivity.

04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Read the rings to reconstruct the climate events a tree lived through." Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before reading the rings, each team writes one prediction and one reason on the handout: which ring pattern is the clearest sign of a hard year. No reading the sequence yet.

Phase 3 Constraints

0:12 - 0:18

Set the rules: read the rings in order from center to bark, map each ring to a year, cite the exact rings as evidence for every inference, and an inference with no cited ring does not score.

Phase 4 Read and infer

0:18 - 0:44

Teams read the ring sequence from center to bark, map each pattern to a year on the answer board, and infer the likely events. Circulate, ask which rings support each call, and resist reading it for them.

Phase 5 Cite the evidence

For each inference, teams fill a claim-evidence card naming the exact rings (which years, what pattern) that justify the call.

Phase 6 Synthesize and re-check

Each team combines its inferences into one short climate history for the tree, then re-checks its least certain inference against the rings before scoring.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, then wipe the answer board, return the ring cards and any wood cookies to the bin, and restock blank claim-evidence cards for the next team. No PPE to remove and nothing to wash.

TROUBLESHOOTING

Problem: A team names events without pointing to rings. **Fix:** Have them point to the exact ring that supports each claim before moving on; treat it as an evidence-quality coaching moment.

Problem: A team makes an inference with no cited rings. Fix: Point to a claim-evidence card and ask which exact rings (years and pattern) support the call; an uncited inference does not score, so coach them to anchor every claim to evidence.

Problem: Two teams read the same rings differently. **Fix:** Have them re-read from center to bark and cite the exact ring for each call; dendrochronologists crossdate and compare many trees to agree on dates.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to defend their climate story with the specific rings.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to rings as proxy data.

Tier 2 (building but not reasoning):

- "Which single ring or run of rings most changes your climate story, and why?"

Tier 3 (ready to extend):

- "Cross-date your tree against a second ring-card set: find a shared stress year in both and use it to line up the timelines."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Which pattern was easiest to misread?
- How is a tree ring like and unlike a thermometer?
- What other natural records store past climate?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Correct inferences	40
Evidence justification	20
Speed	20
Final synthesis	20
Total	100

CLEANUP AND SOURCES

Return the ring cards, answer boards, and any wood cookies to the bin, restock blank claim-evidence cards, wipe the dry-erase boards, and collect score slips. There is nothing wet or chemical to dispose of.

SOURCES NOAA NCEI Tree Ring product and paleoclimatology (https://www.ncei.noaa.gov/products/paleoclimatology/tree-ring); NOAA Climate.gov, How tree rings tell time and climate history (https://www.climate.gov/news-features/blogs/beyond-data/how-tree-rings-tell-time-and-climate-history); UCAR Center for Science Education, Tree Rings and Climate (https://scied.ucar.edu/learning-zone/how-climate-works/tree-rings-and-climate).